4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = 2*(L+W) and Area = L*W.

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
d2_96390_Shashidhar>DELIMITER $$
d2_96390_Shashidhar>
d2_96390_Shashidhar>CREATE PROCEDURE RectangleCalc()
    -> BEGIN
    ->     DECLARE L INT DEFAULT 10;  -- length
    ->     DECLARE W INT DEFAULT 5;   -- width
    ->
    ->     DECLARE perimeter INT;
    ->     DECLARE area INT;
    ->
    ->     SET perimeter = 2 * (L + W);
    ->     SET area = L * W;
    ->
    ->     SELECT L AS Length,
    ->            W AS Width,
    ->            perimeter AS Perimeter,
    ->            area AS Area;
    -> END $$
Query OK, 0 rows affected (0.07 sec)

d2_96390_Shashidhar>
d2_96390_Shashidhar>DELIMITER ;
d2_96390_Shashidhar>CALL RectangleCalc();
+--------+-------+-----------+------+
| Length | Width | Perimeter | Area |
+--------+-------+-----------+------+
|     10 |     5 |        30 |   50 |
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

2. 2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
2_96390_Shashidhar>CREATE TEMPORARY TABLE temp_table (
    ->     num INT,
    ->     square_val INT,
    ->     cube_val INT
    -> );
uery OK, 0 rows affected (0.05 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
d2_96390_Shashidhar>CREATE PROCEDURE InsertValues()
    -> BEGIN
    ->     DECLARE num INT DEFAULT 5;  -- assign value
    ->
    ->     INSERT INTO temp_table (num, square_val, cube_val)
    ->     VALUES (
    ->         num,
    ->         num * num,
    ->         num * num * num
    ->     );
    ->
    ->     SELECT * FROM temp_table;
    -> END $$
Query OK, 0 rows affected (0.01 sec)

d2_96390_Shashidhar>
d2_96390_Shashidhar>DELIMITER ;
d2_96390_Shashidhar>CALL InsertValues();
+------+------------+----------+
| num  | square_val | cube_val |
+------+------------+----------+
|    5 |         25 |      125 |
+------+------------+----------+
1 row in set (0.02 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are C= (F-32)

```
d2_96390_Shashidhar>CREATE PROCEDURE TempConvert(
    ->     IN temp_value DOUBLE,
    ->     IN choice VARCHAR(1)   -- 'C' or 'F'
    -> )
    -> BEGIN
    ->     DECLARE result DOUBLE;
    ->
    ->     IF choice = 'C' THEN
    ->         -- Fahrenheit to Celsius
    ->         SET result = (temp_value - 32) * 5 / 9;
    ->         SELECT result AS Celsius;
    ->
    ->     ELSEIF choice = 'F' THEN
    ->         -- Celsius to Fahrenheit
    ->         SET result = (temp_value * 9 / 5) + 32;
    ->         SELECT result AS Fahrenheit;
    ->
    ->     ELSE
    ->         SELECT 'Invalid Choice (Use C or F)' AS Message;
    ->     END IF;
    ->
    -> END $$
```

*5/9 F= 9/5*C + 32

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
d2_96390_Shashidhar>CREATE PROCEDURE ConvertInches(IN total_inches INT)
    -> BEGIN
    ->     DECLARE yards INT;
    ->     DECLARE feet INT;
    ->     DECLARE inches INT;
    ->
    ->     -- 1 yard = 36 inches
    ->     SET yards = total_inches DIV 36;
    ->
    ->     -- remaining inches after yards
    ->     SET total_inches = total_inches MOD 36;
    ->
    ->     -- 1 foot = 12 inches
    ->     SET feet = total_inches DIV 12;
    ->
    ->     -- remaining inches
    ->     SET inches = total_inches MOD 12;
    ->
    ->     SELECT yards AS Yards,
    ->            feet AS Feet,
    ->            inches AS Inches;
    -> END $$
Query OK, 0 rows affected (0.01 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

4. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.
5.
6.

```
2_96390_Shashidhar>CREATE PROCEDURE CheckProduct(
    ->     IN num1 DOUBLE,
    ->     IN num2 DOUBLE
    -> )
    -> BEGIN
    ->     DECLARE product DOUBLE;
    ->
    ->     SET product = num1 * num2;
    ->
    ->     IF product >= 100 THEN
    ->         SELECT product AS Product,
    ->                'Product is greater than or equal to 100' AS Result;
    ->     ELSE
    ->         SELECT product AS Product,
    ->                'Product is less than 100' AS Result;
    ->     END IF;
    ->
    -> END $$
uery OK, 0 rows affected (0.02 sec)

2_96390_Shashidhar>
2_96390_Shashidhar>DELIMITER ;
```

7.

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
2_96390_Shashidhar>CALL CheckProduct(10, 12);
---------+----------------------------------------+
 Product | Result                                 |
---------+----------------------------------------+
     120 | Product is greater than or equal to 100 |
---------+----------------------------------------+
 row in set (0.02 sec)

uery OK, 0 rows affected (0.03 sec)
```